Qijun Zhou (He/His/Him)

Electrical Engineering

Vancouver, British Columbia

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TECHNICAL SKILLS

COMPUTER

- *C, MATLAB, Embedded C, Python, System Verilog, Quartus and Questa, RISC-V, Visual Studio & Visual Studio Code, GIT & GITHUB, HTML & CSS, Figma, Codex Workflow*

HARDWARE

- *Microcontroller, Oscilloscope, Signal Generator, Breadboard, Soldering Iron, Circuit Analysis / Design, De-10 Lite*

ELECTRICAL

- *Multimeters, Basic Circuit tools, Circuit Design / Components*

EDUCATION

University of British Columbia **Expected Graduation: APR 2029**
Bachelor of Applied Science – Electrical Engineering
Co-op: Available for 12 months beginning Sep 2026

TECHNICAL PROJECTS (COURSE)

Field-Tracking Robot (GROUP) **MAR 2026 – APR 2026**

- Developed microcontroller firmware in Embedded C for track following, intersection handling, motor balancing, and real-time serial debugging
- Processed ADC sensor readings from inductive pickup circuits and used PWM motor control to follow a buried guide wire
- Implemented IR remote mode switching, predefined path execution, scan-mode data collection, and UART logging for testing and calibration
- Built a Python GUI for robot control, serial data collection, and heat-map visualization, enabling scan-mode testing and track calibration
- Used Git version control to manage firmware and Python GUI development, including structured commits, project documentation, and GitHub-based code organization

Oven Reflow Control System (GROUP) **JAN 2026 – APR 2026**

- Designed and implemented a temperature measurement, control, and display system for an oven using the DE10-Lite FPGA development board and external circuit components
- Developed embedded control logic on CV-8052 soft-core in 8051 assembly language, including ADC data acquisition, temperature computation, and display handling
- Implemented a finite state machine (FSM) to control oven operating states, including idle, heating, and steady state, based on temperature thresholds
- Deployed system by synthesizing and loading the CV-8052 soft-core (System Verilog) onto the DE10Lite using Intel Quartus

TECHNICAL PROJECTS (PERSONAL)

Transfer Function Visualizer **MAY 2026 – NOW**

- Developing a Python-based transfer function visualizer that supports filter-mode selection, custom transfer-function input, and Bode plot generation
- Built a Tkinter prototype for low-pass, high-pass, and band-pass filters with parameter input, error checking, and automated plot generation
- Used NumPy and Matplotlib to model filter responses based on cutoff frequency, gain, and termination frequency
- Planned a web-based version with filter selection, custom transfer-function input, reference circuit diagrams, and transfer-function explanations
- Applied Git version control to manage source code, track development progress, and maintain GitHub project documentation

Personal Portfolio Website **APR 2026 – MAY 2026**

- Developed a personal portfolio website using HTML and CSS to showcase engineering projects, technical skills, coursework, and interests
- Designed a multi-page website structure with project detail pages, navigation links, and organized content sections
- Deployed the website through Netlify and connected it with a custom domain for public portfolio presentation
- Applied Git version control to manage source code, track design updates, and maintain GitHub-based documentation
- Website can be found here: *https://kenzhou.netlify.app/*

Research on Technological Innovation of Super High-rise Buildings **JUL 2022 – DEC 2022**

- Published an independent peer-reviewed review paper on technological innovation in super high-rise building design and construction
- Reviewed structural systems, advanced materials, and construction technologies including steel
- structures, high-performance concrete, BIM, ultra-high concrete pumping, and GPS/BDS surveying
- Synthesized engineering literature into practical reference points for safety, sustainability, construction efficiency, and material selection
- Paper can be found here *https://doi.org/10.54097/hset.v28i.4100*

VOLUNTEER

UBC Face Drama Club, Vancouver, BC, Canada **NOV 2024 – Now**

Sound Designer

- Designed and Implement QLab-based audio systems for live theatre, including ambient audio, transitions, and original compositions
- Executed real-time sound cue operation during performances to support narrative flow and stage dynamics

OTHER WORK EXPERIENCE

CICC, Shanghai, China **JUL 2024 – AUG 2024**

Financial Trainee

- Analyzed bond market dynamics by evaluating interest rate trends and yield curve changes to inform investment decisions

- Studied venture capital investment models, including early-stage risk–return profiles, and risk hedging strategies through diversification across asset classes
- Developed and executed project-specific investment strategies, monitoring market sentiment and making timely portfolio adjustments

AWARDS

Top 25% of the world in the 2024 Euclid Competition **MAR 2024**

INTERESTS & ACTIVITIES

- *Released music and received musician certification on the NetEase Cloud Music platform in China*
- *Organized a team and participated in the 2024 Shanghai "Xinmin Evening News" Cup High School Football Tournament, reaching top 16 in city*